Group II Row IV, step by step: where $\log \frac{\vee}{k^+} H_4$ comes from, and what is wrong with the d^3N^y term of Eq. (5.12)

1 The short answer

- **Your advisor is right.** The $d^3N^y_{\xi\mathrm{LO}}$ term of Eq. (5.12), *as written*, has an intrinsic problem:
  - the log stands outside the integral, and then the term is identically zero for $\mathbf{k} \neq 0$;
  - it is not manifestly symmetric under amplitude $\leftrightarrow$ conjugate ($y \leftrightarrow y'$, $x \leftrightarrow x'$);
  - "LO at $y' = y$" is not a cross section.
- **You are also right.** BFKL$\otimes$LO needs $+\log \frac{\vee}{k^+} H_4$, and Row IV contains it. It sits inside the DGLAP integral, at its lower end $\xi \to \frac{k^+}{\vee}$ (Sec. 2).
- **How the two fit together.** When $+\log \frac{\vee}{k^+} H_4$ is taken out of Row IV into the BFKL list, the same amount must be subtracted from the DGLAP integral. That subtraction is what the d^3N^y term was meant to be. Written correctly it is

$$\boxed{-2N_c \frac{1-\xi}{\xi} \frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k} \Big[\log \frac{X^2\mu^2}{4}\Big] \;\longrightarrow\; -2N_c \frac{1-\xi}{\xi} \Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big) \frac{d^3N^y_{\mathrm{LO}}}{d^3k}} \tag{1.1}$$

  with $\frac{d^3N^y_{\mathrm{LO}}}{d^3k}$ symmetric (both gluon points at the same z, no phase), Eq. (1.2). After the ξ integral it is $-\big[\log \frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}\big] H_4$.
- **The advisor's "it should match or cancel with H_4"** is exactly true:
  - the subtraction is proportional to H_4;
  - together with the $-(1 - \frac{k^+}{\vee})H_4$ line it gives $-\log \frac{\vee}{k^+} H_4$;
  - together with the BFKL piece $+\log \frac{\vee}{k^+} H_4$ it gives zero, so the full Row IV has no explicit H_4 (Sec. 2).
- **The running coupling** is $\mathrm{LO}|_{\alpha_s \to \alpha_s(X^2)}$, $X = y' - y$, exactly your advisor's $\alpha_s(X^2) = \alpha_s + \frac{\alpha_s^2\beta_0}{4\pi} \ln(X^2\hat{\mu}^2)$ (Step 4).

Notation: $X \equiv y' - y$, $a \equiv \frac{k^+}{\vee}$, $c_0 = 2e^{-\gamma_E}$, $\hat{\mu}^2 = \mu^2 e^{2\gamma_E}/4 = \mu^2/c_0^2$, $d_\perp = 2 - \epsilon$. $\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}[f]$ is your (5.13) with the weight $f(X)$ *inside* $\int_{x,y,x',y'}$; $\frac{d^3N_{\mathrm{LO}}}{d^3k}[f]$ likewise ($\xi = 1$).

$$\frac{d^3N^y_{\mathrm{LO}}}{d^3k} \equiv \frac{1}{(2\pi)^3} \frac{g^2}{2\pi^2 k^+} \int_{x,x',z} \frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2} \big[U^{e'c}(z) - U^{e'c}(x')\big]\big[U^{ec}(z) - U^{ce}(x)\big] \rho^{e'}(x')\rho^e(x), \tag{1.2}$$

$$H_4 = \frac{g^2 N_c}{\pi \mathbf{k}^2} \frac{d^3N^y_{\mathrm{LO}}}{d^3k}. \tag{1.3}$$

2 Step by step

Step 0: the full Row IV (your three lines, before anything is extracted; $\frac{\xi}{(1-\xi)_+}$ is your p^+ prescription, $p^+ = k^+ \frac{1-\xi}{\xi}$):

$$\begin{aligned}
&(\mathrm{L1}) = \frac{g^2N_c}{4\pi^2} \int_a^1 \frac{d\xi}{\xi^2} \Big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\Big] \frac{d^3N_{\xi\mathrm{LO}}}{d^3k}[\mathcal{B}_0], \qquad (\mathrm{L2}) = \frac{g^2N_c}{4\pi^2} \Big[\log \frac{k^+}{\Lambda} + \log\Big(\frac{\vee}{k^+} - 1\Big)\Big] \frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}_0], \\
&(\mathrm{L3}) = \frac{g^2N_c}{4\pi^2} \int_a^1 d\xi \, \frac{1-\xi}{\xi} \frac{d^3N^{\mathrm{lp}}_{\xi\mathrm{LO}}}{d^3k}.
\end{aligned} \tag{2.1}$$

Every term is symmetric under amplitude $\leftrightarrow$ conjugate. There is no H_4 and no d^3N^y.

Step 1: the ϵ-bracket. $d_\perp = 2-\epsilon$ gives exactly $\mathcal{B}_0 = -\frac{2}{\epsilon} - \frac{d_\perp}{2}\ln(X^2\hat{\mu}^2) \equiv \mathcal{B}$. In terms without a pole in front, $\frac{d_\perp}{2} \to 1$.

Step 2: your p^+ prescription $\to$ the standard plus distribution; (L2) cancels except $\log\frac{k^+}{\Lambda}$. Exactly (`g2iv_full/plus_check.py`, 6×10^{-17}):

$$\text{(L1)} = \frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg} - \frac{\beta_0}{2}\delta(1-\xi)\Big]\frac{d^3N_{\xi\text{LO}}}{d^3k}[\mathcal{B}] - \frac{g^2N_c}{4\pi^2}\log\Big(\frac{\vee}{k^+}-1\Big)\frac{d^3N_{\text{LO}}}{d^3k}[\mathcal{B}], \tag{2.2}$$

where $\mathcal{P}_{gg}$ is the standard splitting function, with the standard plus distribution on $[a,1]$ (endpoint $\log(1-a)$ included). Hence

$$\text{(L1)}+\text{(L2)} = \underbrace{\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\frac{d^3N_{\text{LO}}}{d^3k}[\mathcal{B}]}_{\text{JIMWLK}} + \frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg} - \frac{\beta_0}{2}\delta(1-\xi)\Big]\frac{d^3N_{\xi\text{LO}}}{d^3k}[\mathcal{B}]. \tag{2.3}$$

Step 3: DGLAP. The $\mathcal{P}_{gg}$ term is

$$\text{(D)} \equiv -\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,\frac{d^3N_{\xi\text{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat{\mu}^2)\Big]. \tag{2.4}$$

Its pole is the $\overline{MS}$ counterterm of $D_{h/g}$: the region $a\le\zeta\xi\le\zeta\le1$ is that of $\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}D(\zeta)$. After it, $\ln(X^2\hat{\mu}^2)\to\ln(X^2\mu_F^2/c_0^2)$.

Step 4: running coupling (your advisor's formula). The δ term of (2.3) is $+\frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\text{LO}}}{d^3k}\big[\frac{2}{\epsilon}+\ln(X^2\hat{\mu}^2)\big]$. The coupling counterterm of LO removes $\frac{2}{\epsilon}$, and $\frac{g^2}{16\pi^2}=\frac{\alpha_s}{4\pi}$, LO $\propto\alpha_s$:

$$\frac{d^3N_{\text{LO}}}{d^3k}\Big|_{\alpha_s} + \frac{\alpha_s\beta_0}{4\pi}\frac{d^3N_{\text{LO}}}{d^3k}\big[\ln(X^2\hat{\mu}^2)\big] = \frac{d^3N_{\text{LO}}}{d^3k}\Big|_{\alpha_s\to\alpha_s(X^2)}, \qquad \alpha_s(X^2)=\alpha_s+\frac{\alpha_s^2\beta_0}{4\pi}\ln(X^2\hat{\mu}^2) \tag{2.5}$$

(sympy: difference 0, `advisor/checks.py`).

Step 5: the logarithm hidden in (D). Look at (D) at small ξ. There $\mathcal{P}_{gg}\to 2N_c\frac{1-\xi}{\xi}$, and the y' integral is a Fourier transform at the large momentum $\mathbf{k}/\xi$:

$$I(\xi)\equiv\int d^2y'\,e^{-ik\cdot(y'-y)/\xi}\,\ln(X^2\hat{\mu}^2)\,F(y') = -\frac{4\pi\xi^2}{\mathbf{k}^2}F(y)\Big[1+O(\xi)\Big], \tag{2.6}$$

where $F(y')$ is the smooth rest of the integrand (kernel, Wilson lines, charges, after $\int_{x'}$).

- The constant and $\ln\hat{\mu}^2$ multiply $(2\pi\xi)^2\delta^2(\mathbf{k})=0$.
- $\nabla^2\ln X^2 = 4\pi\delta^2(X)$ gives $-4\pi\xi^2/\mathbf{k}^2$.
- At $y'=y$: $\frac{(x'-y')}{(x'-y')^2}\to\frac{(x'-y)}{(x'-y)^2}$ and $U(y')\to U(y)$, i.e. $F(y)$ is the integrand of $\frac{d^3N^y_{\text{LO}}}{d^3k}$, with both gluon points at one point ($y\equiv z$).

Numerical check (`smallxi.py`, smooth model F, $|\mathbf{k}|=2$): $I(\xi)\big/\big(-\frac{4\pi\xi^2}{\mathbf{k}^2}F(y)\big)$:

ξ	0.5	0.2	0.1	0.05	0.03
ratio	$1.196-0.712i$	$1.008-0.215i$	$1.002-0.104i$	$0.99994-0.048i$	$0.99979-0.034i$
predicted $O(\xi)$ ($-i\xi$ here)	$-0.5i$	$-0.2i$	$-0.1i$	$-0.05i$	$-0.03i$

So, with $\frac{d\xi}{\xi^2}\cdot\frac{1-\xi}{\xi}\cdot\xi^2=\frac{(1-\xi)d\xi}{\xi}$, the small-$\xi$ end of (D) is

$$-\frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\,2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\frac{d^3N^y_{\text{LO}}}{d^3k} = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N^y_{\text{LO}}}{d^3k}\Big[\log\frac{\vee}{k^+}-(1-a)\Big] = \Big[\log\frac{\vee}{k^+}-\Big(1-\frac{k^+}{\vee}\Big)\Big]H_4. \tag{2.7}$$

This is where $+\log\frac{\vee}{k^+}H_4$ comes from: the parent gluon with very large p^+ (BFKL region) inside the DGLAP integral.

Step 6: H_4 is the same object. Your H_4 line has $\int_w e^{ik\cdot w}\frac{(w-z)^i}{(w-z)^2}=\frac{2\pi ik^i}{\mathbf{k}^2}e^{ik\cdot z}$ (checked), so $\int_{w,w'}(\dots)=\frac{4\pi^2}{\mathbf{k}^2}$ and $H_4=\frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N^y_{\text{LO}}}{d^3k}$, Eq. (1.2). Expanding its colour factor, $\log\frac{\vee}{k^+}H_4$ is exactly your four-line BFKL term (prefactor $2N_c\frac{g^4}{16\pi^5}=\frac{g^4N_c}{8\pi^5}$).

Step 7: extraction (add and subtract). Write (D) as

$$(\mathrm{D}) = \underbrace{-\frac{g^2}{8\pi^2}\int_a^1 \frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat{\mu}^2)\Big] - 2N_c\frac{1-\xi}{\xi}\Big(-\frac{4\pi\xi^2}{\mathbf{k}^2}\Big)\frac{d^3N^y_{\mathrm{LO}}}{d^3k}\Big\}}_{(\mathrm{D_{sub}})\text{: DGLAP, BFKL region subtracted}}$$
$$+\underbrace{\log\frac{\vee}{k^+}H_4}_{\to\ \mathrm{BFKL}} - \Big(1-\frac{k^+}{\vee}\Big)H_4. \tag{2.8}$$

- In ($\mathrm{D_{sub}}$) the integrand is integrable at $\xi\to 0$, by Eq. (2.6) (difference $O(\xi)$ relative). So ($\mathrm{D_{sub}}$) has no log $\frac{\vee}{k^+}$.
- The subtraction has no pole and no μ, so the DGLAP renormalization (Step 3) is unchanged.
- Every term is symmetric.

Step 8: Row IV, final.

$$\begin{aligned}\text{Row IV} = {} & \underbrace{\frac{g^2N_c}{4\pi^2}\log\frac{k^+}{\Lambda}\frac{d^3N_{\mathrm{LO}}}{d^3k}[\mathcal{B}]}_{\mathrm{JIMWLK}} + \underbrace{\log\frac{\vee}{k^+}H_4}_{\mathrm{BFKL}} \\ & - \frac{g^2}{8\pi^2}\int_a^1\frac{d\xi}{\xi^2}\Big\{\mathcal{P}_{gg}\frac{d^3N_{\xi\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat{\mu}^2)\Big] + \frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2}\frac{d^3N^y_{\mathrm{LO}}}{d^3k}\Big\};\ (\mathrm{DGLAP_{sub}}) \\ & + \frac{g^2\beta_0}{16\pi^2}\frac{d^3N_{\mathrm{LO}}}{d^3k}\Big[\frac{2}{\epsilon}+\ln(X^2\hat{\mu}^2)\Big] \quad \text{(running coupling)} \\ & - \Big(1-\frac{k^+}{\vee}\Big)H_4 + \frac{g^2N_c}{4\pi^2}\int_a^1 d\xi\,\frac{1-\xi}{\xi}\frac{d^3N^{\mathrm{lp}}_{\xi\mathrm{LO}}}{d^3k}\Big|_{d_\perp=2} \quad \text{(finite)}\end{aligned} \tag{2.9}$$

(the second term in the braces is Eq. (1.1): $-2N_c\frac{1-\xi}{\xi}\cdot(-\frac{4\pi\xi^2}{\mathbf{k}^2}) = \frac{8\pi N_c\xi(1-\xi)}{\mathbf{k}^2}$).

- **Check:** adding the BFKL term back, the three H_4 structures cancel:

$$\log\tfrac{\vee}{k^+}H_4 \underbrace{-\big[\log\tfrac{\vee}{k^+}-(1-a)\big]H_4}_{\text{subtraction}} -(1-a)H_4 = 0,$$

  and (2.9) returns to Steps 0–4. This is your advisor's "it should cancel with H_4".

- **Correction to Eq. (5.12)**: replace

$$-2N_c\frac{1-\xi}{\xi}\frac{d^3N^y_{\xi\mathrm{LO}}}{d^3k}\Big[\log\frac{(y'-y)^2\mu^2}{4}\Big] \quad\text{by}\quad +\frac{8\pi N_c\,\xi(1-\xi)}{\mathbf{k}^2}\frac{d^3N^y_{\mathrm{LO}}}{d^3k},$$

  and drop (5.14). The value is the same, but it is now well defined and symmetric. Also put the brackets of the DGLAP and RC lines inside the integrals.

- At leading power ($a\to 0$): $\int_a^1\to\int_0^1$, $(1-a)H_4\to H_4$.

3 Checks

p^+ prescription = standard plus $-F(1)\log(\frac{\vee}{k^+}-1)$; the cut $p^+\geq\Lambda$ = your (L2)	6×10^{-17}, 4×10^{-16} (`g2iv_full/plus_check.py`)
RC = $\alpha_s(X^2)$ (advisor's formula)	exact (`advisor/checks.py`)
small-ξ limit (2.6), ratio $\to 1$, next order $-i\xi$	table above (`smallxi.py`)
$\int_w e^{ik\cdot w}\frac{(w-z)^i}{(w-z)^2} = \frac{2\pi ik^i}{\mathbf{k}^2}e^{ik\cdot z}$; $\int d^2r\,e^{-iq\cdot r}\ln r^2 = -\frac{4\pi}{q^2}$	4.83322 vs 4.83322; $-7.45\to-7.44$
$H_4 = \frac{g^2N_c}{\pi\mathbf{k}^2}\frac{d^3N^y_{\mathrm{LO}}}{d^3k}$; $\int_a^1\frac{1-\xi}{\xi} = \log\frac{1}{a}-(1-a)$	exact